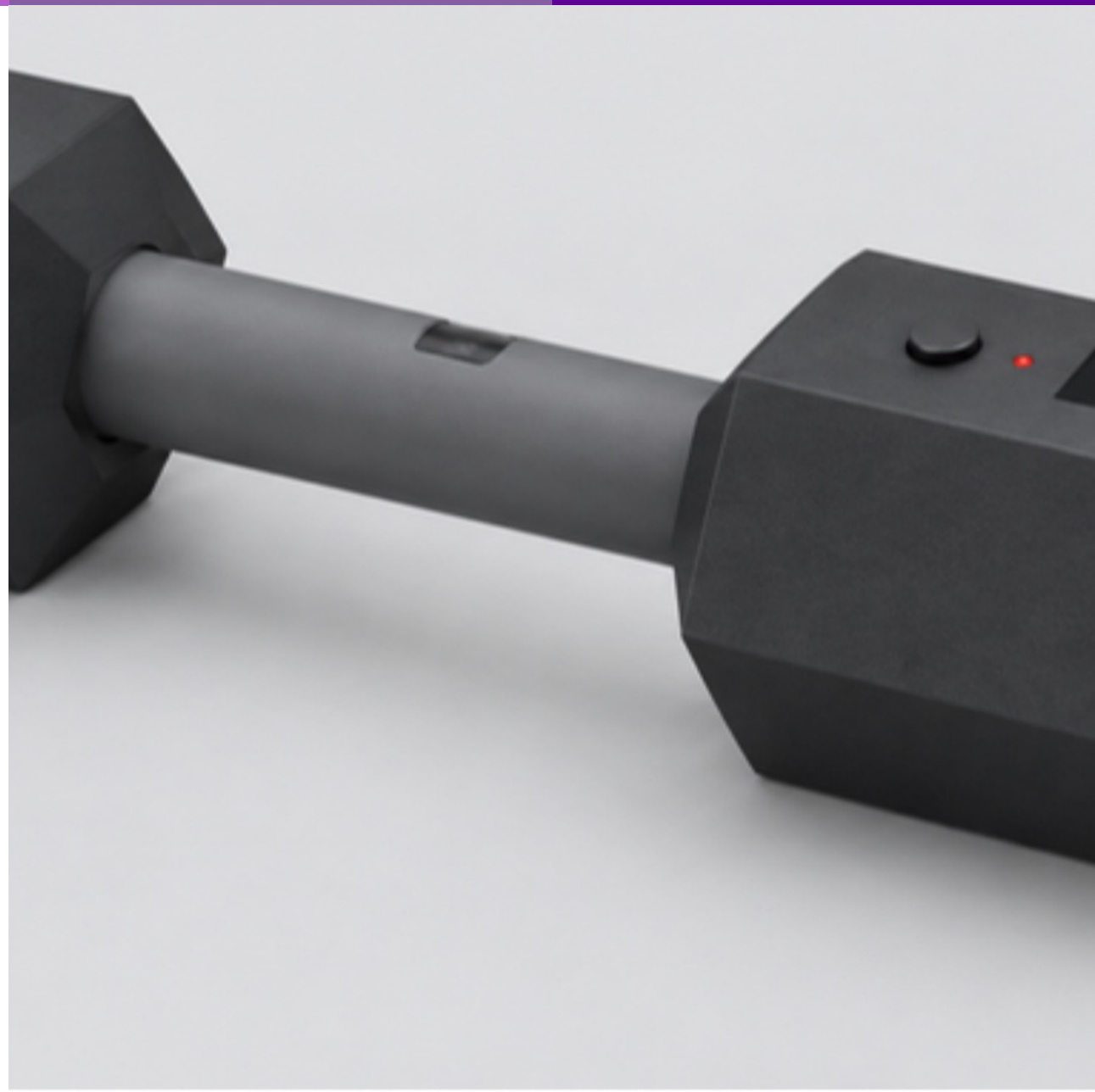
SMART DUMBBELL

Project update, 10 November 2026

ECE 198 Group 57: Hanson He, Eric Zhu, Zoe Wang, Julia Zhang

Counts bicep curls on the device, reads heart rate at the grip, logs every set to a laptop



UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

THREE WEEKS IN: EVERY MODULE RUNS, THE SYSTEM IS HALF INTEGRATED

WORKS ON THE BENCH

- Nano ESP32 + IMU + pulse sensor + OLED on one I2C bus, 4 devices, no conflicts
- IMU streams 6 axes at 104 Hz, axes verified against the silkscreen
- Wi-Fi access point answers HTTP from a phone
- Integrated firmware v0: one button, display, 100 Hz pulse sampling, on-device model call
- Laptop logger prototype with CSV export (on a mock server)
- Schematic v2 drawn in EasyEDA, reviewed; Bluetooth web dashboard merged

IN PROGRESS THIS WEEK

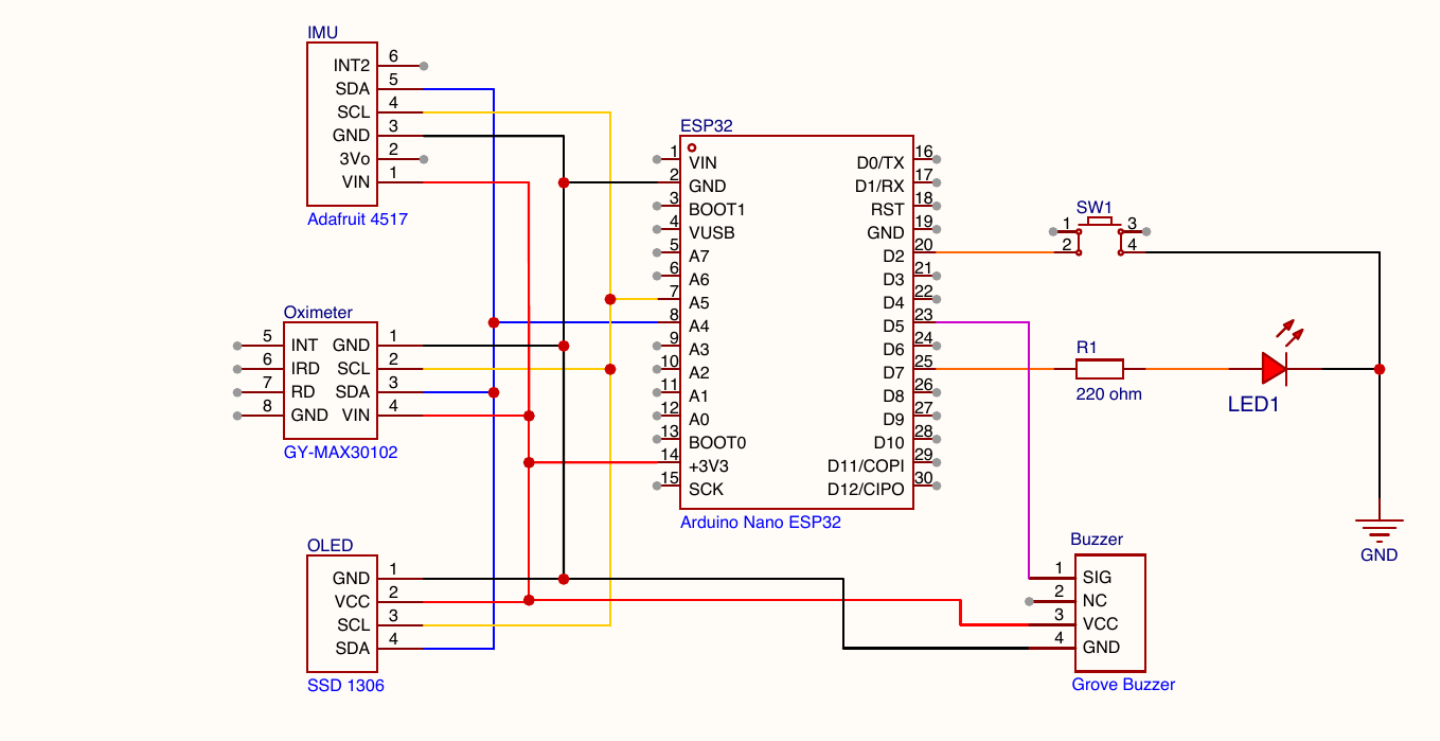
- Rewiring the bench onto a half-size breadboard after the first board failed
- Rep counter and heart-rate detector are placeholders in the integrated app; both need real-motion tuning
- Recording rig and protocol ready; first 3 people this week
- Enclosure layout v0.2 drawn; first print this week

NOT STARTED

- Model trained on real recordings (pipeline tested on synthetic data only)
- Device-side Bluetooth service for the dashboard; logger against real firmware
- Any of the ten acceptance tests
- Printed enclosure

Parts arrived 3 Nov · bring-up records in tests/results · integrated firmware first run 6 Nov

THE BENCH PROTOTYPE, MEASURED ON 3 AND 6 NOVEMBER



Check	Result	Status
I2C devices found	4 of 4	pass
IMU sample rate	104.0 Hz	pass
Samples lost with display on	0 in 20 s	pass
Pulse sampling	100 Hz	pass
Model inference on device	2.3 ms	pass
Button start / pause / end	events logged	pass
Rep count on a real curl	—	not tested
Heart-rate value	9–14 bpm high	open

Found and fixed: the display library dropped the bus to 100 kHz and a full-frame refresh delayed IMU samples; the frame is now sent in 64-byte chunks between samples.

Schematic v2 (EasyEDA, 8 Nov) · one button on D2 · all modules on the Nano 3V3 rail · records: tests/results, 3 Nov and 6 Nov

COUNTING CURLS: A CLASSIFIER GATES A SIMPLE COUNTER

IMU 104 Hz

3-axis accel + 3-axis gyro, X axis along the grip

2 s window, 31 features

mean, spread, range, zero crossings per axis; one result per second

Classifier on the Nano

logistic regression: curl / idle / other with confidence; trained on the laptop, exported as a C header

Counter (placeholder)

gyro rate crosses +60 then -60 deg/s and settles, 0.6-5 s per rep; gate on the curl label once the model is real

WHY TWO PARTS

The counter alone counts any swing. The classifier alone cannot tell one rep from two. Together: a half rep, a pause or picking the dumbbell up does not count (FR-05, FR-06).

WHERE IT STANDS

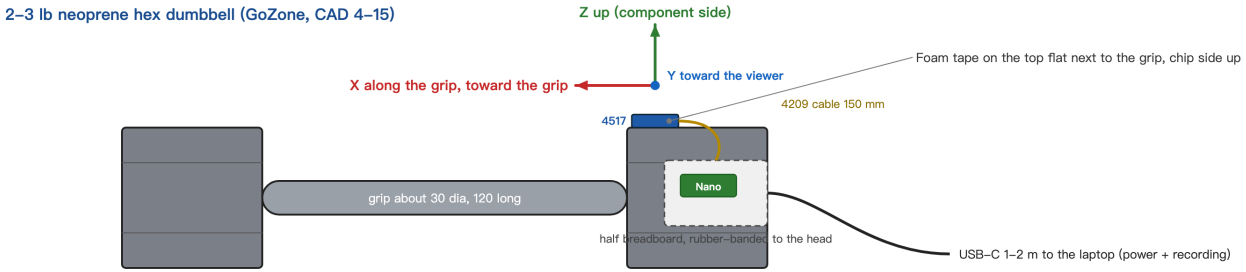
Training pipeline and on-device inference agree to 1e-6, but only on synthetic data. The counter thresholds are guesses and have not been tested on a real curl. Real recordings start this week; evaluation is split by person, never by random windows. Target: macro-F1 ≥ 0.85, count within ±1 on 18 of 20 sets.

ml/workplan.md · classification: Hanson · counting and integration: Julia

TRAINING DATA STARTS THIS WEEK, BEFORE THE ENCLOSURE EXISTS

Temporary rig for recording IMU training data, v1 (until the enclosure exists)

Only the Nano and the 4517 move with the dumbbell; one USB-C cable goes to the laptop. 4517 orientation, fixed once chosen: X along the grip toward the grip, component side up when the dumbbell rests flat.



Mass: dumbbell 0.9–1.4 kg + boards and cables about 80 g, matching the product target. Flat hex faces: tape holds, it does not roll, Z is the same every time.

Rules

1. The 4517 must not move: foam tape plus a turn of tape over it. A loose sensor teaches the model its own rattle.
2. Photograph the mounting as ml/figures/mounting-v1.jpg, write it into ml/data-protocol.md section 3, then never change it. The enclosure mounts the 4517 the same way so the data stays valid.
3. Only the Nano on the breadboard; the four 4209 leads go into the holes beside its pins: 3V3, GND, A4, A5. No OLED, no pulse sensor.
4. Before recording: 5 s at rest shows az about +9.8 and gx gy gz about 0 (Z up); two curls show gx far above gy and gz (X along the grip).

RIG

IMU taped to a 3 lb hex dumbbell, X axis along the grip, orientation photographed and frozen; the same mounting goes into the printed head.

PROTOCOL

6 people, anonymous IDs. Per person: curls slow / normal / fast, three kinds of idle, two minutes of other movements, one minute of half reps. 104 Hz CSV over USB.

WEIGHT

Main recordings at 1.4 kg, checks at 0.9 and 2.3 kg. Product target 1.5 kg, washer ballast adjustable 1–3 kg: senior strength classes use 1–3 lb, early cardiac rehab 1–5 lb.

ml/data-protocol.md · evaluation split by person, never by random windows · 0 recordings as of 8 Nov

HEART RATE AT THE GRIP IS OUR BIGGEST OPEN RISK

0.1–0.4 %

pulse amplitude relative to DC, finger still, at rest

+9 to +14

bpm error of our best offline estimate vs a watch reading 86, at rest

- Sensor, bus and finger detection work; the library beat detector does not. A band-pass plus peak-picking replacement is planned, not yet on the board.
- A curl is about 1 Hz, in the same band as heart rate, so motion will make this worse.
- Firmware already follows FR-08: no finger, too few beats or stale data shows HR -- and logs invalid, never an old value.

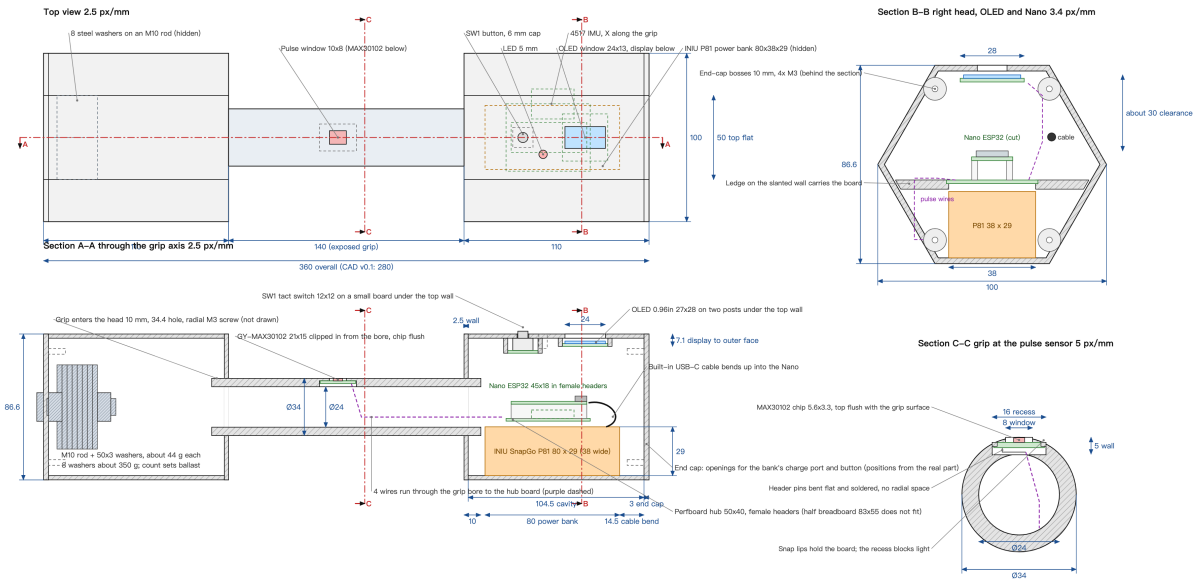
NEXT TWO WEEKS

- Replace the detector with band-pass filtering and peak picking
- Borrow a reference device and run the 60 s synchronized comparison (AT-04)
- Decide before the method freeze: keep valid heart rate during motion, or report it only when signal quality allows and guarantee it between sets

We would like your view on that spec change today.

Measurements 3 Nov, tests/results/2026-11-03_bringup-05-max30102.md · outcome recorded as inconclusive

ENCLOSURE, THE ROAD TO 25 NOVEMBER, AND THREE QUESTIONS



Layout v0.2: electronics and the 5000 mAh power bank in one hex head, steel-washer ballast in the other, pulse sensor flush in the grip. Target 1.5 kg, adjustable 1–3 kg by washer count. First print this week.

Week	Milestone
10–16 Nov	Recordings from 4+ people; first real model; counter tuned offline; printed head fit check; logger talks to the real firmware
17–21 Nov	Model on device gating the counter; assembled dumbbell; count, interference, 10 min and reconnect tests on the device
22–24 Nov	Freeze, spare board flashed, three rehearsals of the 90 s demo
25 Nov	Symposium: start, 10 counted curls, swing not counted, grip heart rate, set logged on the laptop

QUESTIONS FOR YOU

1. May we restate the in-motion heart-rate requirement as above?
2. Symposium format: table, power, how long per visitor?
3. We are at CAD 120 of our own money; is a 200 ceiling reasonable?

Enclosure: Eric · electronics and heart rate: Zoe · classification: Hanson · counting, integration, logger: Julia

QUESTIONS?

WHAT WE WOULD LIKE YOUR VIEW ON

1. Heart rate during motion: keep FR-07 as written, or report it only when signal quality allows and guarantee it between sets?
2. Symposium on 25 November: table, power, time per visitor?
3. Budget: CAD 120 of our own money spent; is a 200 ceiling reasonable?

Repository: github.com/juliazhang1015/smart-dumbbell